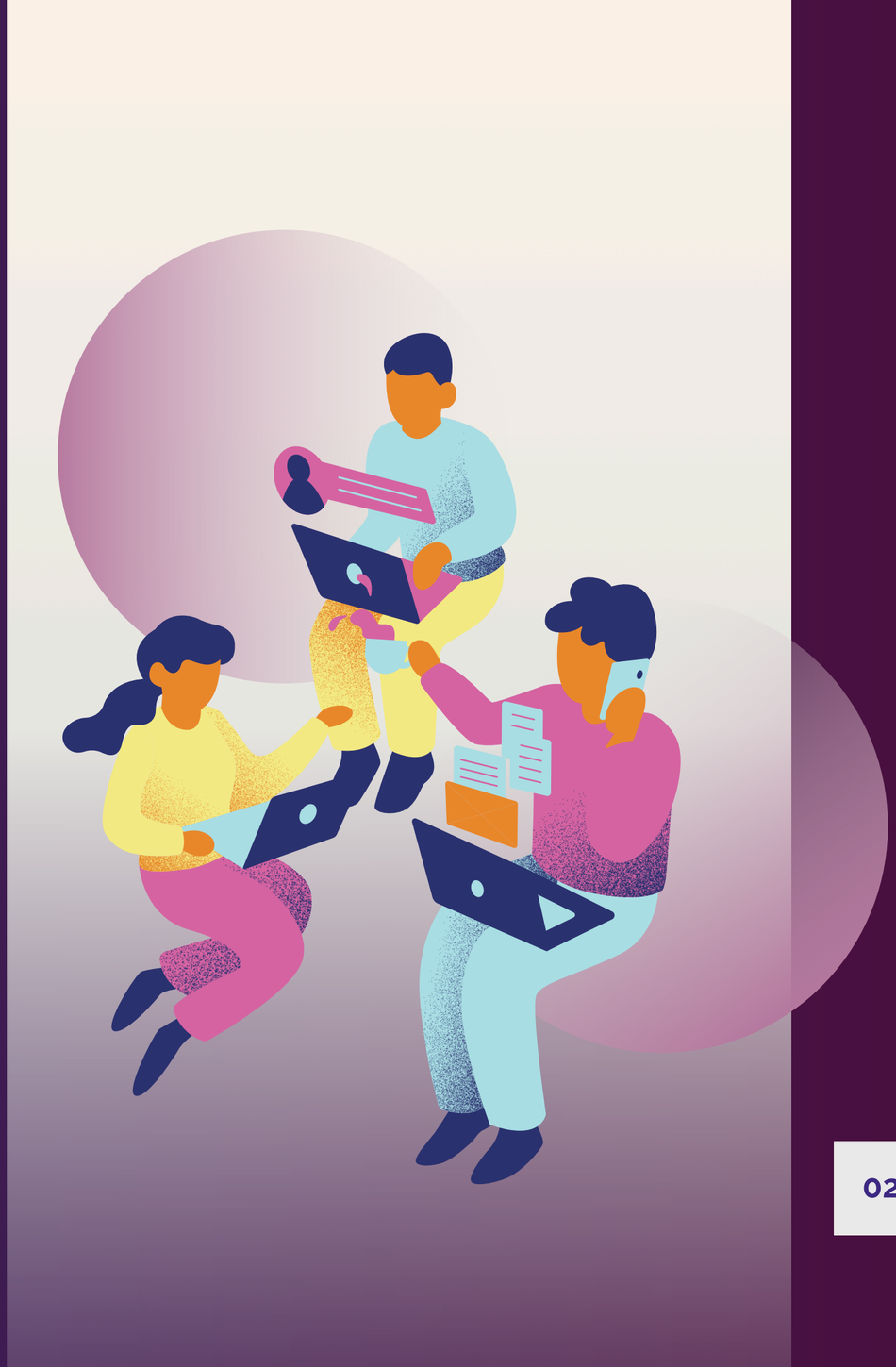
DESIGN AND IMPLEMENTATION OF AN ENTERPRISE DATA INTELLIGENCE PLATFORM FOR UNIFIED ANALYTICS



PROJECT AREA: DATABASE SYSTEMS AND APPLICATIONS
FUNCTIONAL DOMAIN: CONSUMER GOODS AND RETAIL ANALYTICS

PROBLEM STATEMENT

Retail organizations generate data across sales, inventory, procurement, supply chain, customer management and finance.
Data is distributed across heterogeneous systems and formats.
Fragmented data leads to:
Inconsistent reporting
Limited cross-functional visibility
Delayed decision-making
Traditional BI often requires SQL knowledge and predefined reporting workflows.



PROPOSED SOLUTION

- Build a unified analytical environment for enterprise retail data.
- Enable business-user self-service analytics through semantic AI.
- Improve query accuracy, usability and decision-making efficiency.

PROPOSED ARCHITECTURE

BUSINESS INTELLIGENCE

Visible data analysis and dashboards



INTELLIGENCE & ANALYTICS

Advanced AI and semantic analysis

DATA INTEGRATION & STORAGE

Centralized cloud data lake

DATA WAREHOUSE

Structured storage for analytical data

DATA SOURCES

Diverse origins of raw data

EXPECTED IMPACT AND CONTRIBUTION

Business Impact

- Unified analytics across business functions
- Faster and more informed decision-making
- Reduced manual reporting effort
- Improved accessibility and consistency of data
- Better cross-functional visibility

Technical Impact

- Scalable cloud-native architecture
- Improved data quality and governance
- Metadata management & data lineage
- Incremental data processing
- Query and warehouse performance optimization
- Modular and extensible architecture

EXPECTED IMPACT AND CONTRIBUTION

Research Contribution

- Evaluate semantic AI vs. traditional BI
- Assess natural-language analytics for business users
- Establish a reference architecture for enterprise self-service analytics

Key Outcome

A unified, scalable and AI-enabled analytics platform that bridges enterprise data engineering with business-user self-service analytics.



TOOLS AND TECHNOLOGIES

- DATA SOURCES
  - REST API
  - EXCEL/CSV
  - PARQUET
- AZURE ACCOUNT BLOB STORAGE
- SNOWFLAKE
  - DATABASE
  - COWORK

THANK YOU!

DATA ANALYSIS IS KEY TO BUSINESS GROWTH AND SUCCESS!